Exno: 7
Date: 13/8/25

Flow Control at Data link layer using Sliding window protocol

Aim:

To write a program to implement flow control at the Data link layer using the sliding window protocol, and to simulate the flow of frames from one to another.

Algorithm:

1. Start the program.
2. Read the total number of frames (N), window size (w) and lost frame (optional)
3. Initialize base = 0 and next_seq = 0
4. While base < N :
   * Set the sender window as [base, base + w - 1]
   * Send frames within the window and increase next seq.
   * If a frame is lost.
   ① Receiver send it and later frames.
   ② Sender time out and retransmits from the lost frames.
   * otherwise :
   ① Receiver sends ack for received frames.
   ② Sender slides the window.
5. Repeat until all frames are transmitted and acknowledge.
6. Stop the program.

Program:

```
def simulate(n, w, lost):
    base = 0
    next_seq = 0
    while base < n:
        end = min(base + w - 1, n - 1)
        print("n ---------- ")
        print(f" Sender Window : [Frame {b} - Frame {w}]")
        while next_seq < b + w and next_seq < n:
            print(f"Node A -> Sending Frame {next_seq}")
            if next_seq = lost:
                print(f" (Frame {next_seq} LOST in transit)")
            next = se + = 1
            if lost is not None and b <= lost < next_seq:
```

```
        print(f"Node A -->Timeout ! Resending window starting from frame {lost}.")
        next_seq = lost
        lost = None
        continue
    for i in range(b, next_seq):
        print(f"Node B --> Received frame {i}, sending ACK({i})")
    print(f"Node A --> ACK {next_seq - 1} received. Sliding window forward.\n")
    b = next_seq
    print("\n ----------")
    print(f"All {n} frames successfully transmitted from Node A to Node B.")

def main():
    print("\n=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION
                    (Go-Back-N) ===\n")
    n = int(input("Enter total number of frames to send : "))
    w = int(input("Enter sender's window size : "))
    lost = int(input("Enter frame number to simulate as lost in (0 to {n-1}),
                enter -1 for no frame): "))
    if lost == -1:
        lost = None
    simulate(n, w, lost)

if __name__ == "__main__":
    main()
```

Results:

The program to implement flow control at the Data link layer using the sliding window protocol was successfully written in Python, executed and verified by simulating the flow of frames between two nodes.

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send : 8
Enter sender's window size : 3
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss): 4

-----------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

-----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
    (Frame 4 LOST in transit!)
Node A --> Sending Frame 5
Node B --> Frame 4 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 4.

-----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received. Sliding window forward.

-----------------------------------------
Sender Window : [Frame 6 - Frame 7]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

-----------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```